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Myungjin Lee (she/her)

M.S.E. in Computer Science & Artificial Intelligence

Email: leemj637@gmail.com

Personal Website: https://myungjin00.github.io

GitHub: https://github.com/myungjin00

Interests

Graph Representation Learning, Molecular Property Prediction, Drug Repurposing, Explainable AI (XAI)

Education

M.S.E. in Computer Science and Artificial Intelligence Mar 2024 – Aug 2026
Dongguk University, Seoul, Republic of Korea
- Advisor: Dr. Sangsoo Lim
- Dissertation: *Graph Substructure-Based Junction Tree Modeling for Drug ADMET Prediction*
- Major: AI in Healthcare and Medicine
- GPA 4.04/4.5

B.E. in Industrial & Management Engineering Mar 2023 – Feb 2024
(Advanced Major Program)
Induk University, Seoul, Republic of Korea
- GPA 4.43/4.5
- Valedictorian

A.E. in Industrial & Management Engineering Mar 2020 – Feb 2023
Induk University, Seoul, Republic of Korea
- GPA 4.41/4.5 (Major: 4.42/4.5)
- Valedictorian

Research Experience

Dongguk Univ. PRISM Lab, Seoul, Republic of Korea Mar 2024 – Aug 2026
- *Position:* Researcher (Advisor: Dr. Sangsoo Lim)
- *Role:* Molecular graph representation learning and explainable AI for drug discovery — interpretable ADMET prediction from molecular substructures (first-author SJoINT; co-author MAGNET); a drug-repurposing model developed and registered to the national bio-data platform (KBDI); and bulk/scRNA-seq omics analysis for target discovery.

Manuscripts

(In preparation) **Lee M.**, Mo J., Kang M., Lim S*, SJoINT: Substructure-Driven Junction Tree for Interpretable ADMET Prediction
- Dual-view learning of atom-level graphs and substructure-level junction trees via structure-constrained bidirectional cross-attention with augmentation-free contrastive pretraining (ZINC250K); evaluated on MoleculeNet ADMET and MoleculeACE activity-cliff tasks.

(Under review, Bioinformatics) Mo J., **Lee M.**, Lee S., Lim S*, MAGNET: Cross-view Molecular Graph Learning for Interpretable ADMET Prediction
- Multi-view molecular graph framework integrating BRICS, junction-tree, and Murcko fragmentations via cross-view meta-graphs, with self-supervised pretraining on ZINC250K and ChemBERTa-enhanced fine-tuning across 10 MoleculeNet ADMET benchmarks.

Poster Presentations

Korea Computer Congress (KCC) 2026, Jeju, Korea
- *Structure-Constrained Bidirectional Cross-Attention for Self-Supervised Molecular Representation Learning*

BIOINFO 2025, Suwon, Korea
- *Weighted Junction-Tree Nodes for Enhanced Interpretability in ADMET Tasks*

Grants

Master's Student Research Grant — PI Sep 2024 – Aug 2025
(석사과정생 연구장려금지원사업)
- *Ministry of Science and ICT (MSIT).* Individual research grant on functional-group junction-tree-based multi-task molecular graph representation learning.
(Advisor: Dr. Sangsoo Lim)

Research Projects *(Participating researcher)*

멀티모달 AI 기반 노화연관 지방간질환(MASLD) 표적 발굴 및 약물 검증 *(RS-2025-00560523)* Mar 2025 – Feb 2026
PI: Dr. Sangsoo Lim (Dongguk University);
Collaborator: Dr. Gung Lee (Mayo Clinic, USA).
- *Role:* Preliminary omics/bioinformatics for target discovery: curated public & experimental bulk/single-cell RNA-seq; QC → DEG → GSEA/ssGSEA → module scoring; characterized lipid-metabolism reprogramming and cellular-senescence signatures (SenMayo/SASP) in aging mouse liver, with a supporting cross-check in public human data; prioritized candidate genes and Plin2-defined senescent-cell subsets.

개방형 AI 신약개발·데이터 분석 플랫폼 개발센터 (RS-2025-18732993) Jul 2025 – Aug 2026
PI: Dr. Minho Lee (Dongguk University)
- *Role:* (**Drug repurposing)** built a standardized PrimeKG-based benchmark to evaluate five models (TxGNN, DREAMWalk, DRHGCN, AMDGT, SVGA) across 190 diseases / 3,379 drugs; developed DC-VGAE and a performance-based ensemble. (**ADMET)** benchmarked and developed an interpretable ADMET model using junction-tree / Murcko / BRICS substructures with cross-attention. **(Deployment)** packaged the drug-repurposing model as a Docker image and registered it to the national bio-data platform (KBDI).

Challenge

2024 Samsung AI Challenge — Machine Learning Force Field Aug 2024
- Samsung Electronics SAIT · DACON
- **Top 10% (11th)**
- Built MLFF models predicting atomic energies and forces with uncertainty quantification for semiconductor-material simulation (team DAI).

DREAM Olfactory Mixtures Prediction Challenge Aug 2024
- International DREAM Challenge, Sage Bionetworks
- Graph neural network predicts olfactory similarity of molecular mixtures.

Honors and Awards

- **Highest Academic Achievement Award** 2023

Induk University, Korea.

- **Merit-based Academic Scholarship —awarded every semester** 2020 – 2023

Induk University, Korea (incl. GRAPE Talent Scholarship).

Patents and SW Registration	**Software Registration** ▪ (No. C-2025-040207) An Embedding Method via Multi-view Graph Integration of Molecular Structures, Korea Copyright Commission, 2025 ▪ (No. C-2024-042597) Deep Learning Model for Similarity Prediction of Graph-based Mixtures, Korea Copyright Commission, 2024 **Korean Patent Application (pending)** ▪ (No. 10-2025-0003146) System and Method for Predicting Olfactory Characteristics of Odor Mixture Data, 2025	
Technical Skills	**Programming**	Python, R (basic), Java, C#, JavaScript, SQL
	Deep Learning	PyTorch, PyTorch Geometric, Hugging Face Transformers, scikit-learn
	Data Analysis & Visualization	pandas, NumPy, anndata, seaborn, matplotlib
	Cheminformatics	RDKit, DeepChem, TorchDrug, ChemPy
	Bioinformatics	Scanpy, Seurat, bulk & scRNA-seq, differential expression (DESeq2 / PyDESeq2, MAST), GSEA / ssGSEA, GO enrichment
	Tools & Databases	Git, Linux, Docker, MySQL
Language	Fluent in **English** and Native in **Korean** ▪ TOEIC 860 (LC 440 / RC 420)	